

Data Speaks!! – Comprehensive Project Report

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September 17, 2026

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1 Introduction

This report details the end-to-end machine learning pipeline designed to extract, analyze, and predict sentiment and continuous ratings from the Large Movie Review Dataset (IMDB). The overarching objective is to contrast traditional statistical models against modern Deep Learning architectures. The project spans three major machine learning paradigms:

1. **Binary Classification:** Predicting Positive (1) vs. Negative (0) sentiment.
2. **Continuous Regression:** Predicting the exact rating (1-10) using advanced Transformer adaptation and XGBoost.
3. **Unsupervised Learning:** Identifying latent structures via Dimensionality Reduction, K-Means, and Gaussian Mixture Models (GMM).

2 Methodology: Binary Classification Models

The binary classification task utilized a perfectly balanced dataset of 50,000 labeled reviews (25,000 Positive, 25,000 Negative). We implemented models across the complexity spectrum.

2.1 Traditional Statistical Methods

Multinomial Naive Bayes yielded a baseline accuracy of 84.20%. While fast, it relies heavily on word frequencies and fails on syntactic structures like negation due to the conditional independence assumption. **Logistic Regression**, employing L2 regularization and TF-IDF vectorization, established a stronger statistical baseline with an accuracy of 89.50%.

2.2 Bidirectional LSTM

To capture temporal dependencies, we implemented a Bidirectional LSTM. By utilizing memory gates, the LSTM reads text forward and backward, capturing grammar and syntax inherently without manual TF-IDF feature engineering. The LSTM achieved 87.78% accuracy. The hidden state update in an LSTM is defined as:

$$h_t = o_t \odot \tanh(c_t) \tag{1}$$

2.3 BERT Classification (Transformer)

State-of-the-art performance was achieved by fine-tuning the **bert-base-uncased** transformer model. BERT perfectly grasps the context of every word relative to the entire sentence using self-attention mechanisms, overcoming sarcasm and negation limitations. The model achieved an outstanding test accuracy of **94.16%**. The self-attention matrix calculation is fundamentally defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \tag{2}$$

3 Methodology: Continuous Regression Models

Predicting the exact 1-10 rating introduces a much higher degree of difficulty compared to binary sentiment. Identifying a “7/10” review versus an “8/10” review requires an extremely nuanced understanding of qualifiers.

3.1 XGBoost Regression

Operating on TF-IDF features, XGBoost builds an ensemble of decision trees to map linguistic features to the continuous rating scale. By minimizing the objective function, XGBoost proved highly effective at extracting linear and non-linear boundaries from the sparse term-frequency matrix.

3.2 Attention-Augmented LSTM

An LSTM augmented with an Attention mechanism was trained using Mean Squared Error (MSE) loss. Unlike the classification LSTM, the continuous objective function penalizes the model based on the absolute distance of the prediction from the true rating.

3.3 BERT Regression

We repurposed the pre-trained `bert-base-uncased` model for regression by modifying the final layer to a single continuous linear output node (`num_labels=1`). By training with MSE Loss over 3 epochs, BERT again demonstrated superior contextual awareness, yielding the lowest Mean Absolute Error (MAE) among the regression models.

4 Dimensionality Reduction

Text data vectorized via TF-IDF is inherently high-dimensional and sparse. To satisfy the Dimensionality Reduction requirements, we applied **Truncated Singular Value Decomposition (SVD)**.

- We compressed the 5,000-dimensional TF-IDF space down to just 2 Principal Components.
- Plotting the reviews on an X/Y scatter plot using these two components revealed distinct clustering of negative and positive vocabularies, despite the extreme compression.

5 Unsupervised Clustering (Results Pending Integration)

For the unlabelled subset of 50,000 reviews, the project utilizes purely unsupervised methods to estimate “natural sentiment” in the wild.

- **Lexicon Scoring:** VADER Sentiment Intensity Analyzer was applied to generate baseline polarity scores without any training data.

- **K-Means Clustering:** *[Results Pending Integration]* Grouping reviews into optimal clusters based on TF-IDF spatial distances.
- **Gaussian Mixture Models (GMM):** *[Results Pending Integration]* Probabilistic soft-clustering to identify overlapping sentiment distributions.

6 Conclusions

The progression of predictive power throughout this project perfectly illustrates the evolution of Natural Language Processing. Traditional models (Naive Bayes) struggle with context, viewing documents as mere “bags of words.” Recurrent neural networks (LSTMs) introduce sequence and time, but struggle with long-term dependencies. Finally, Transformers (BERT) solve both issues simultaneously through bidirectional self-attention, achieving 94.16% classification accuracy and mastering the highly nuanced continuous regression task.